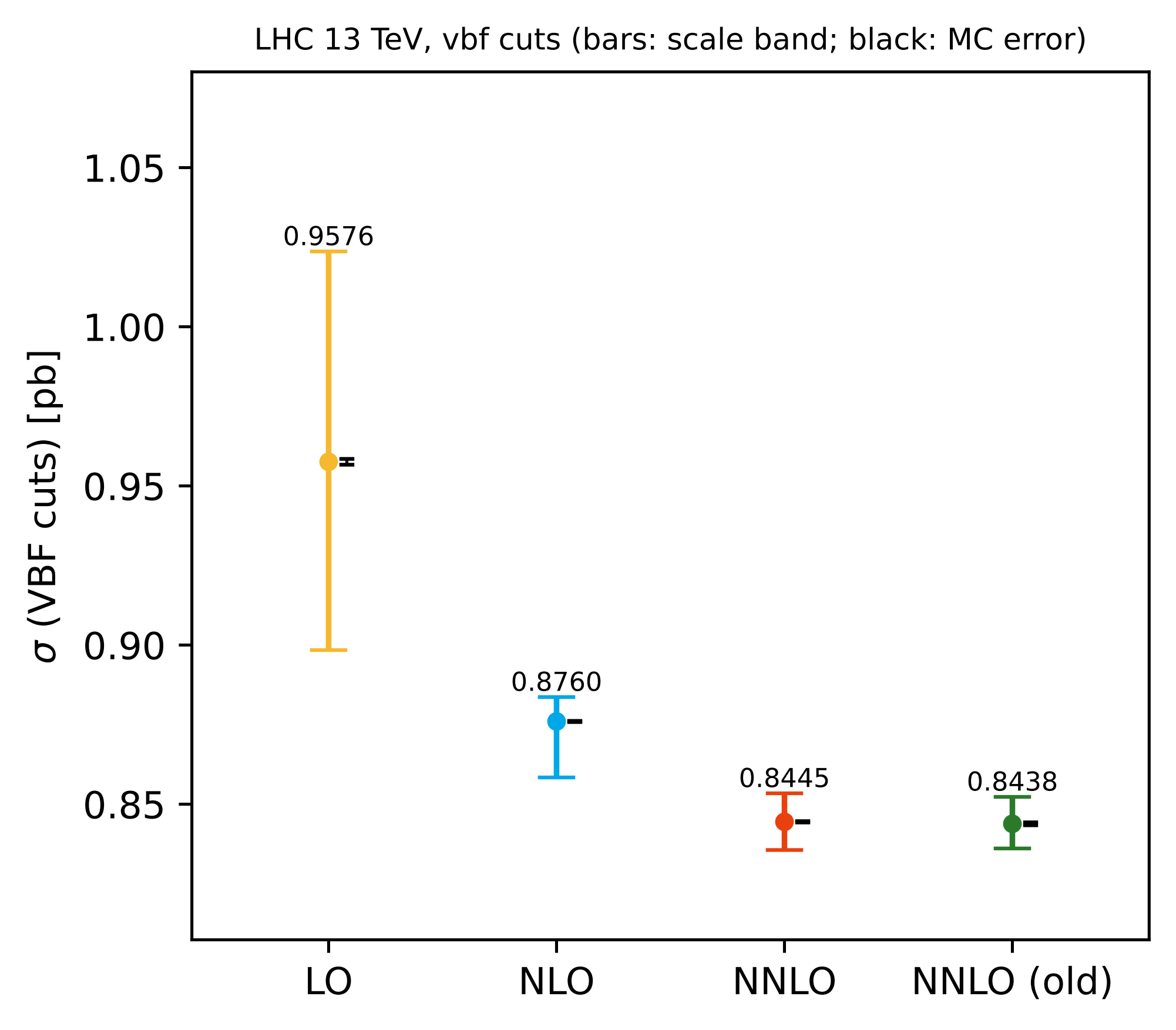
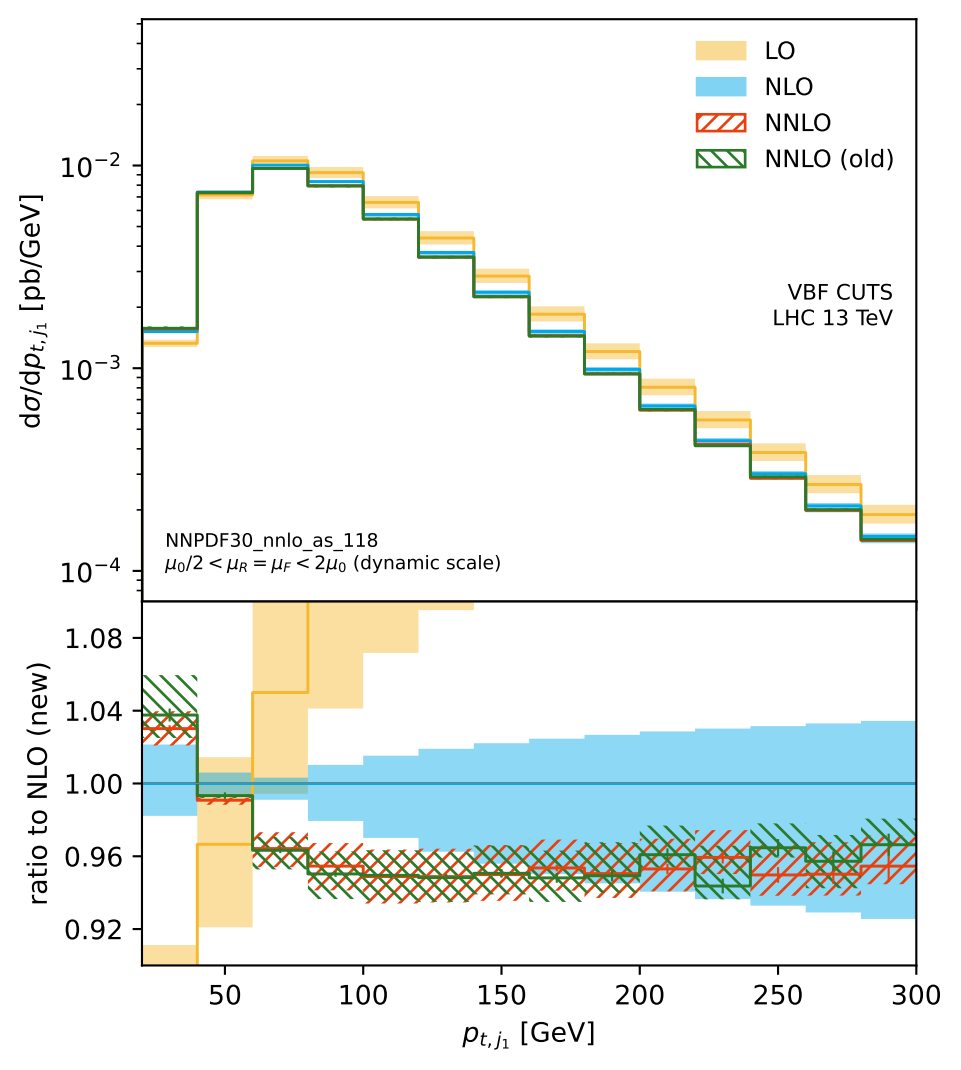
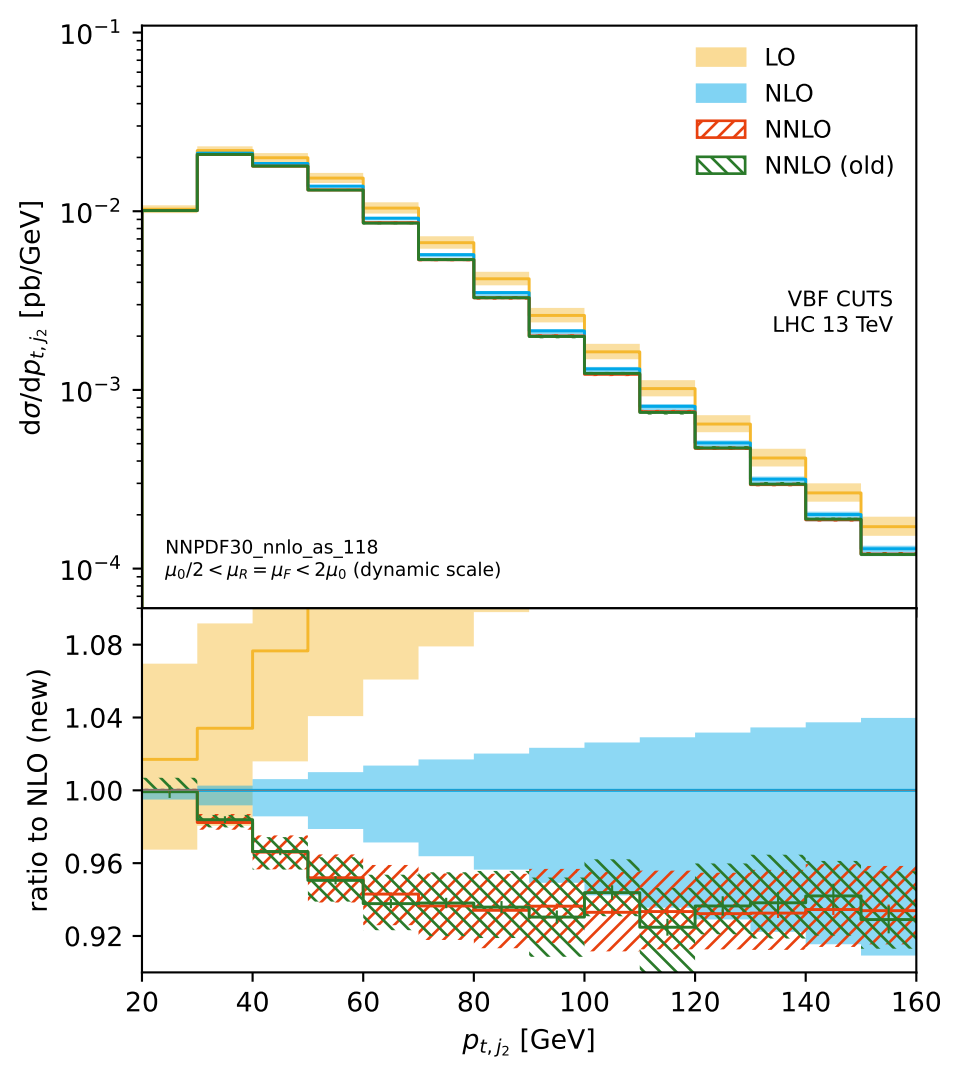
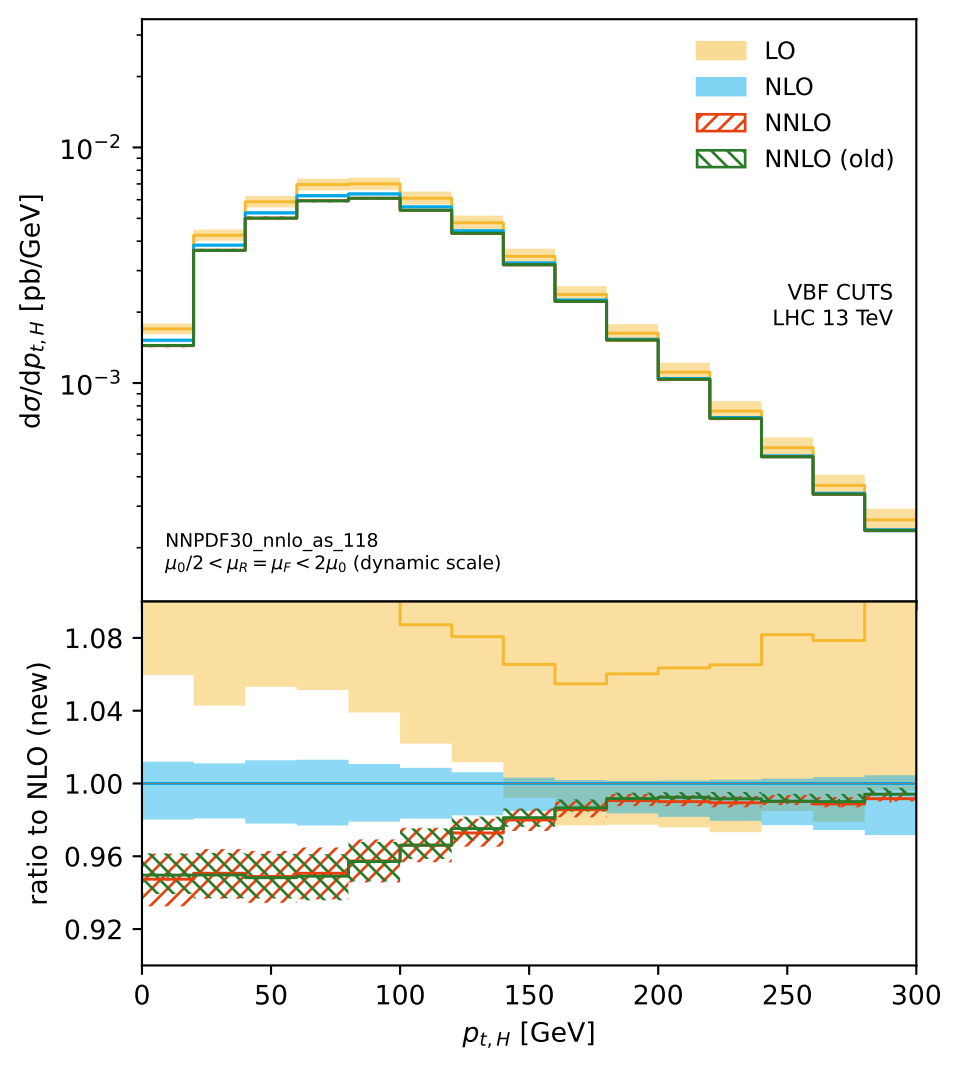
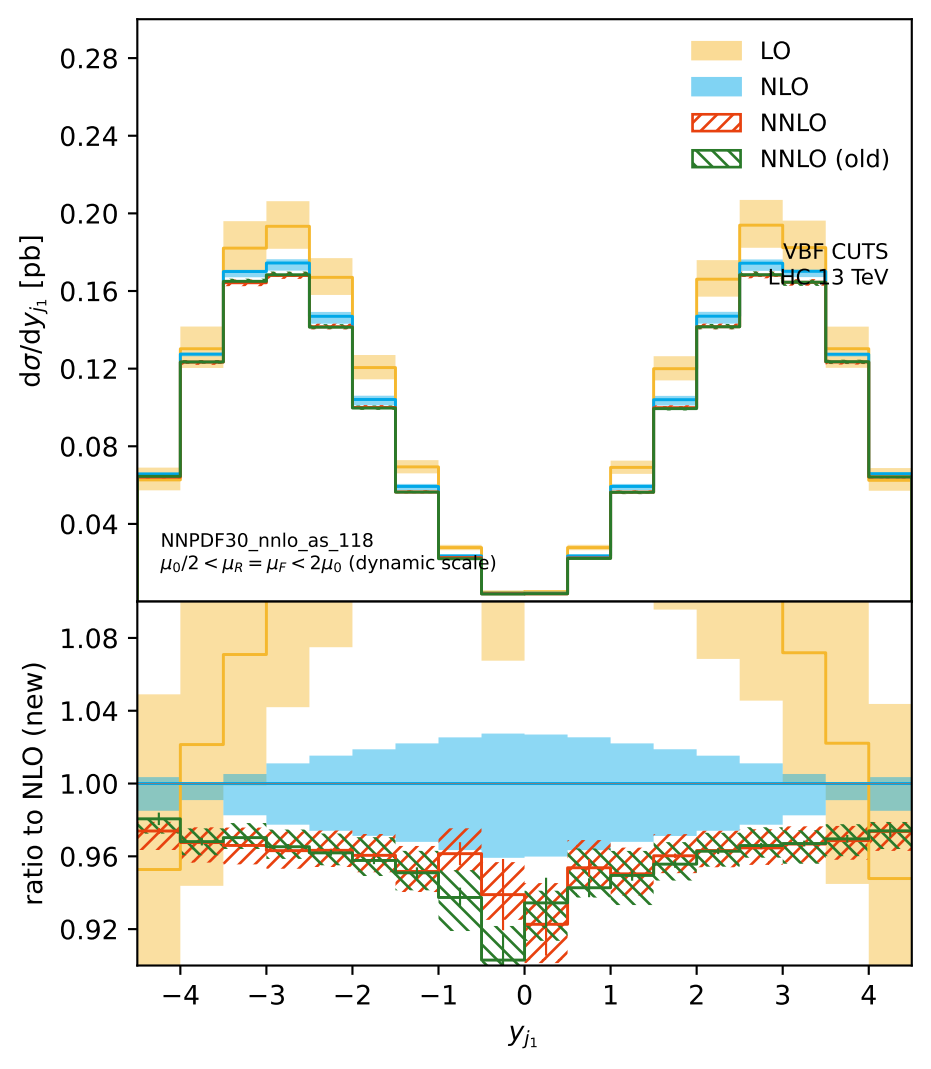
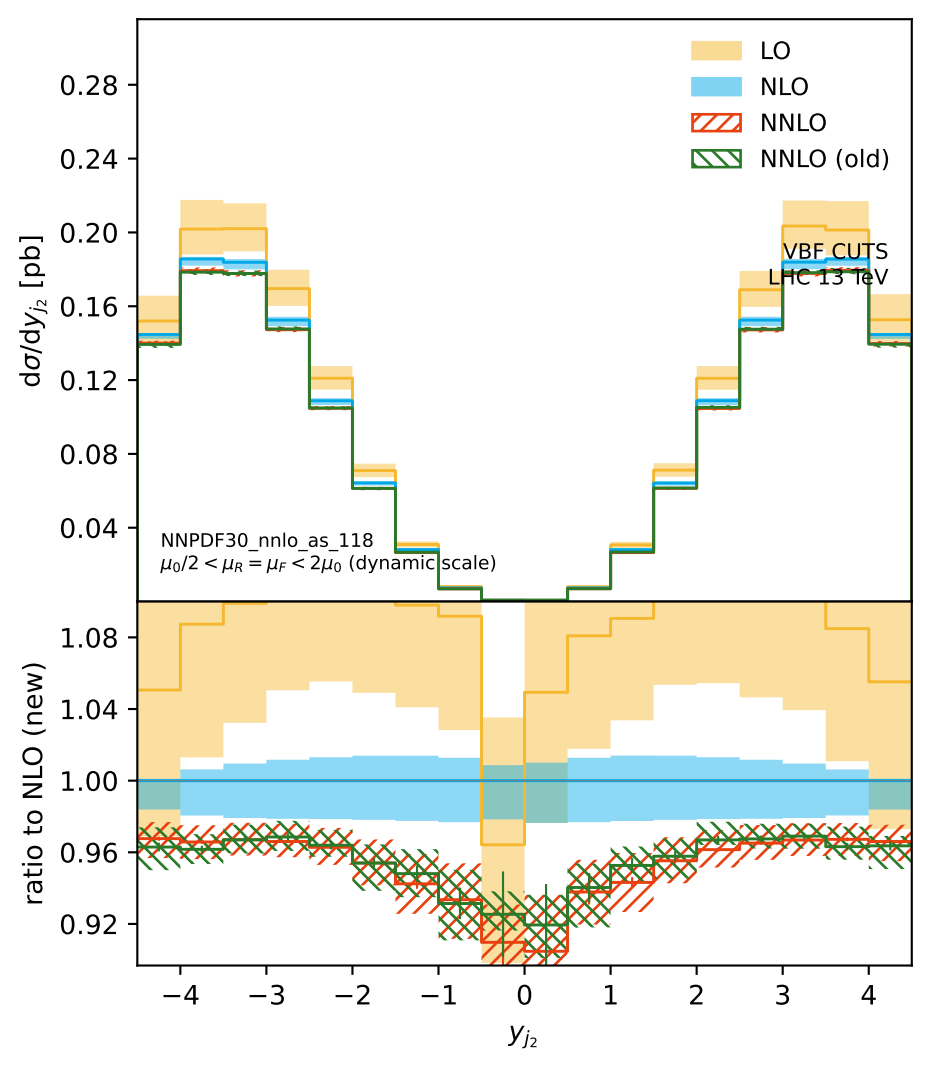

LHC 13 TeV, vbf cuts (bars: scale band; black: MC error)
σ (VBF cuts) [pb]
1.05
1.00
0.95
0.90
0.85
0.9576
0.8760
0.8445
0.8438
LO
NLO
NNLO
NNLO (old)

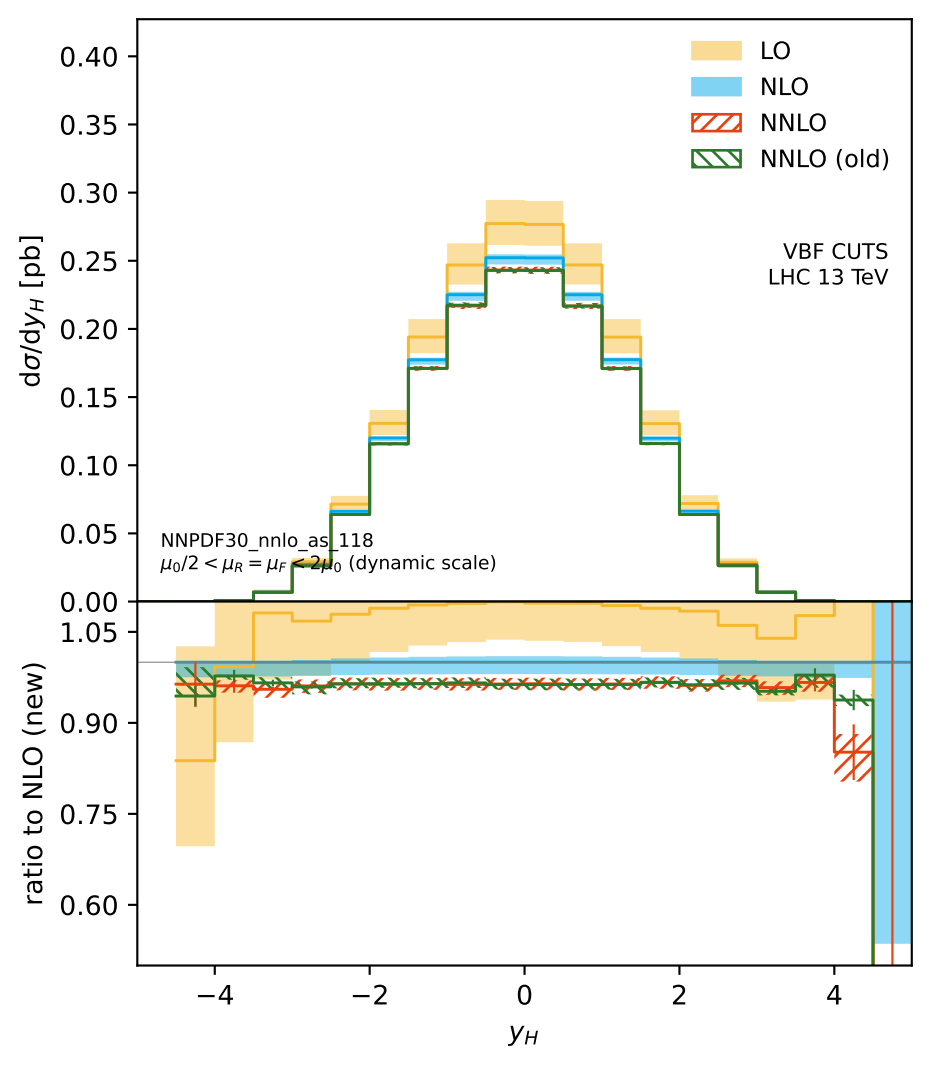

LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dp_{t,j_1}$ [pb/GeV]
10^{-2}
10^{-3}
10^{-4}
ratio to NLO (new)
1.08
1.04
1.00
0.96
0.92
50
100
150
200
250
300
p_{t,j_1} [GeV]

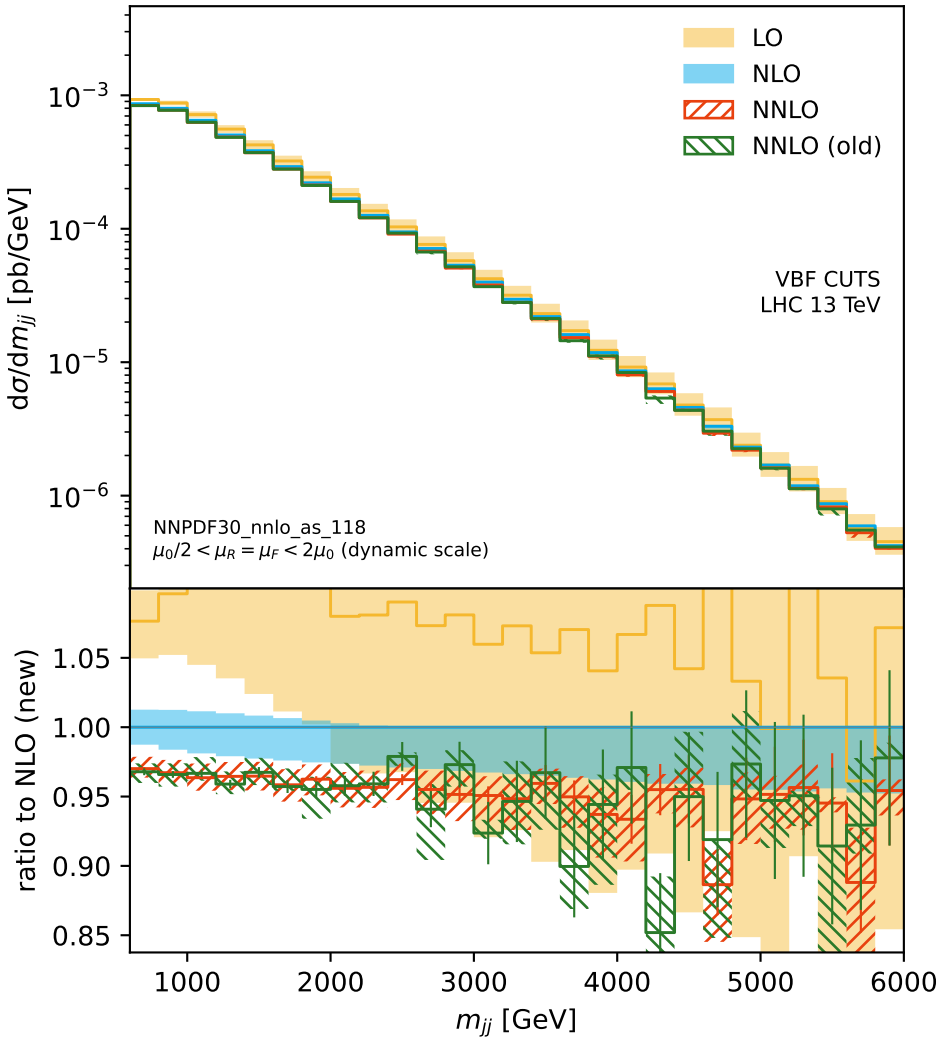

LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dp_{t,j_2}$ [pb/GeV]
ratio to NLO (new)
p_{t,j_2} [GeV]

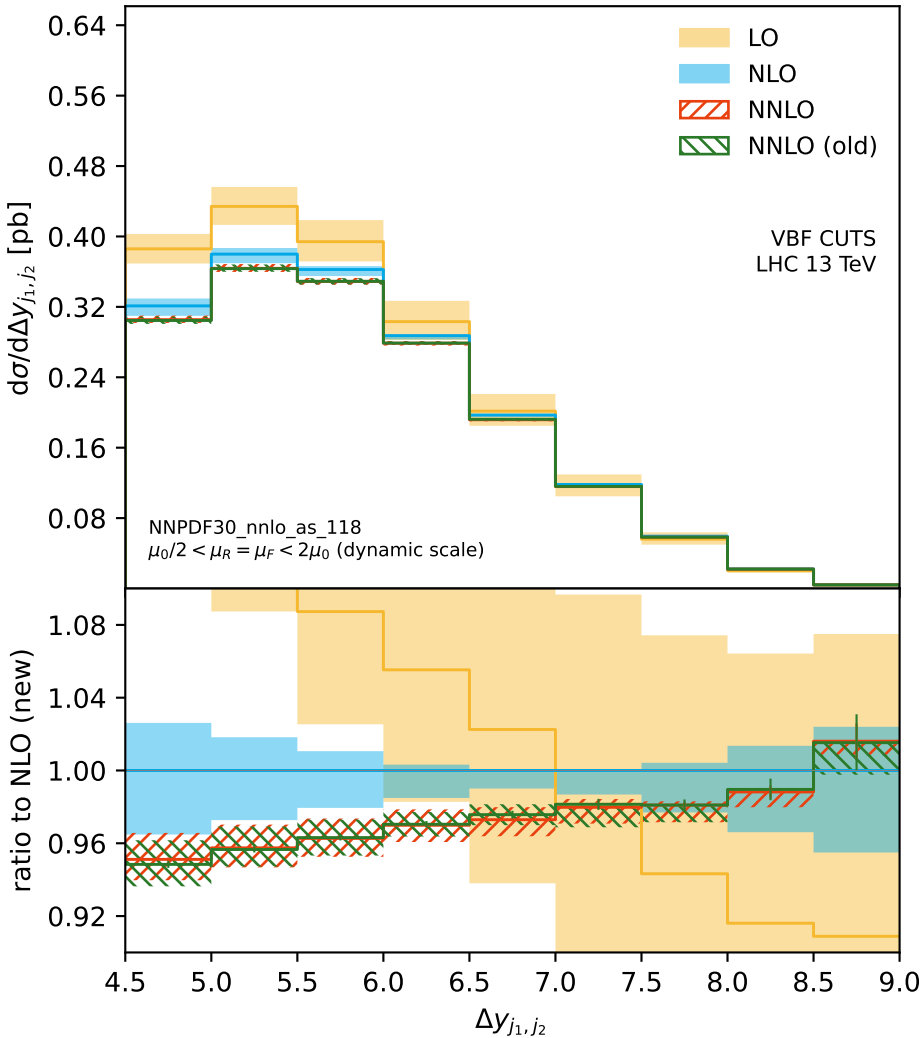

LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dp_{t,H}$ [pb/GeV]
10^{-2}
10^{-3}
ratio to NLO (new)
1.08
1.04
1.00
0.96
0.92
0
50
100
150
200
250
300
$p_{t,H}$ [GeV]

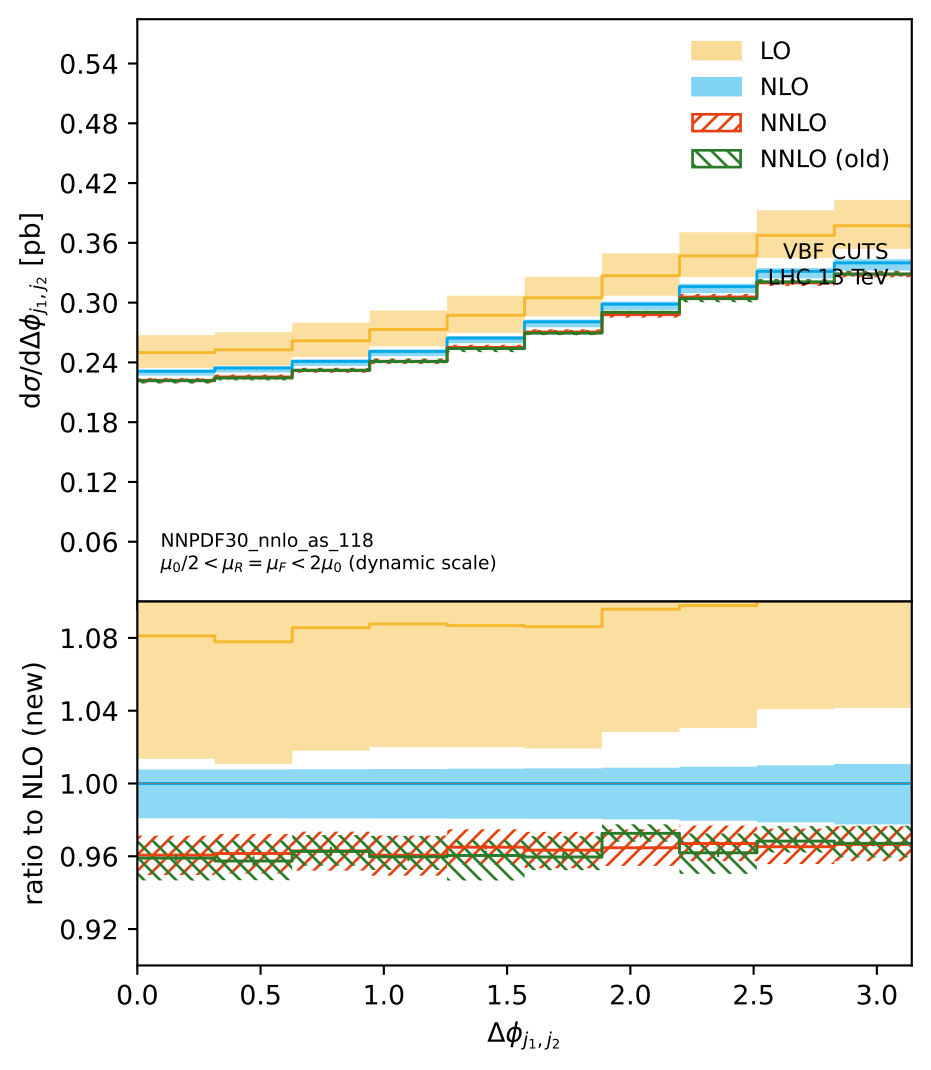

LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dy_{j_1}$ [pb]
ratio to NLO (new)
y_{j_1}
0.28
0.24
0.20
0.16
0.12
0.08
0.04
1.08
1.04
1.00
0.96
0.92
−4
−3
−2
−1
0
1
2
3
4

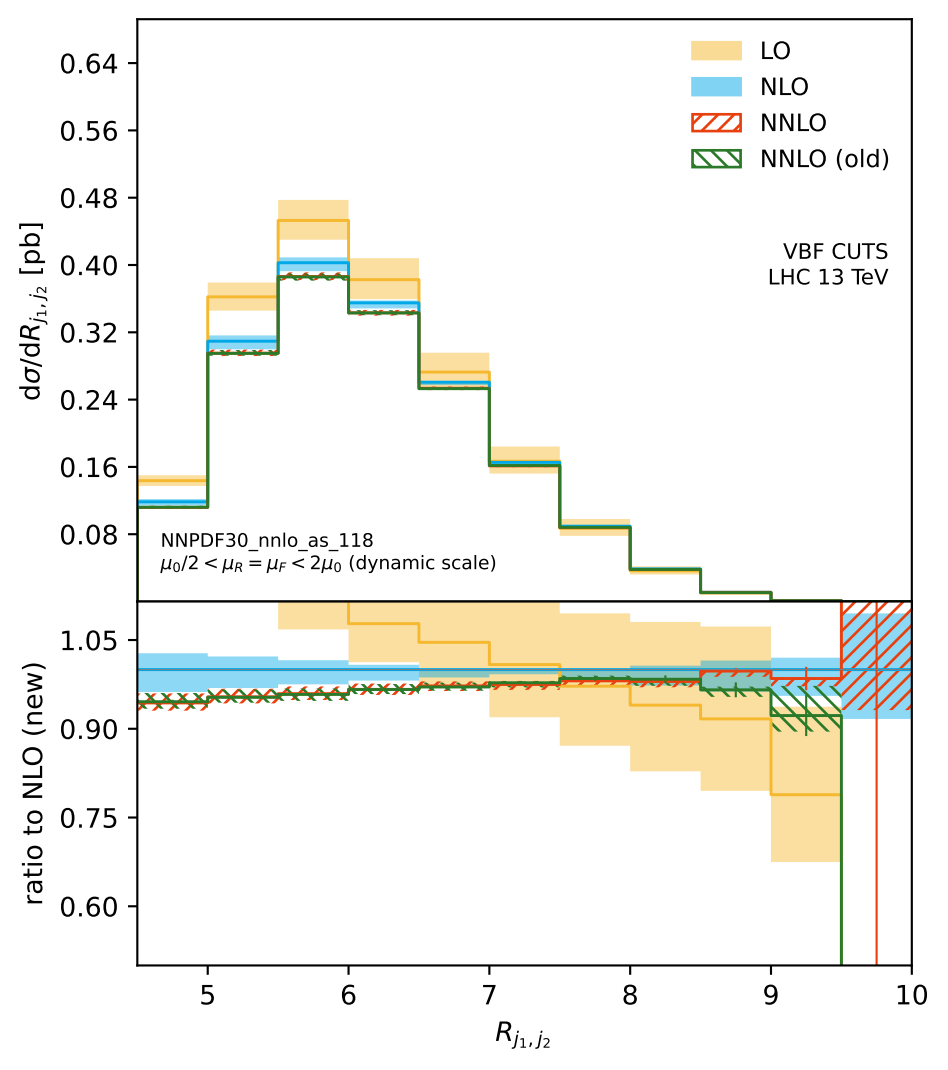

LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dy_{j_2}$ [pb]
0.28
0.24
0.20
0.16
0.12
0.08
0.04
ratio to NLO (new)
1.08
1.04
1.00
0.96
0.92
−4
−3
−2
−1
0
1
2
3
4
y_{j_2}


LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dy_H$ [pb]
ratio to NLO (new)
y_H
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.05
0.90
0.75
0.60
−4
−2
0
2
4


LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dm_{jj}$ [pb/GeV]
10^{-3}
10^{-4}
10^{-5}
10^{-6}
ratio to NLO (new)
1.05
1.00
0.95
0.90
0.85
1000
2000
3000
4000
5000
6000
m_{jj} [GeV]


LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/d\Delta y_{j_1,j_2}$ [pb]
ratio to NLO (new)
$\Delta y_{j_1,j_2}$
0.64
0.56
0.48
0.40
0.32
0.24
0.16
0.08
1.08
1.04
1.00
0.96
0.92
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
8.5
9.0


LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/d\Delta\phi_{j_1,j_2}$ [pb]
ratio to NLO (new)
$\Delta\phi_{j_1,j_2}$
0.54
0.48
0.42
0.36
0.30
0.24
0.18
0.12
0.06
1.08
1.04
1.00
0.96
0.92
0.0
0.5
1.0
1.5
2.0
2.5
3.0


LO
NLO
NNLO
NNLO (old)
VBF CUTS
LHC 13 TeV
NNPDF30_nnlo_as_118
$\mu_0/2 < \mu_R = \mu_F < 2\mu_0$ (dynamic scale)
$d\sigma/dR_{j_1,j_2}$ [pb]
ratio to NLO (new)
R_{j_1,j_2}
0.64
0.56
0.48
0.40
0.32
0.24
0.16
0.08
1.05
0.90
0.75
0.60
5
6
7
8
9
10